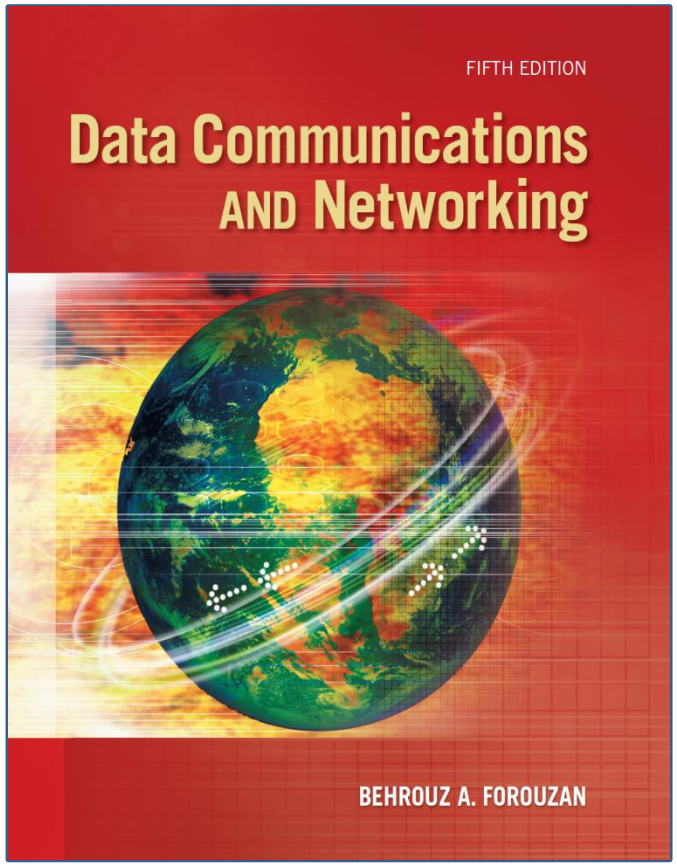


Network Services (Telnet, SSH, and FTP)

Lecturer: Dr. Khaled BinAbdl

Lecture 6

TELNET

- TELNET is **generic client/server** program that **allows** a user on the **client** site to **log into** the computer at the **server** site and **use** the **services** available there.

- For **example**, if a **student** needs to **use** the **Java compiler** program at his **university lab**, there is no need for a Java compiler **client** and a Java compiler **server**.

TELNET

- The **student** can use a client program to **log** into the university **server** and **use** the **compiler** program **at the university**.

- We **refer to** these **generic** client/server programs as ***remote logging*** applications.

- One of the original remote logging protocols is **TELNET,**
  - which is an abbreviation for ***TE****rmina****L NET****work.***

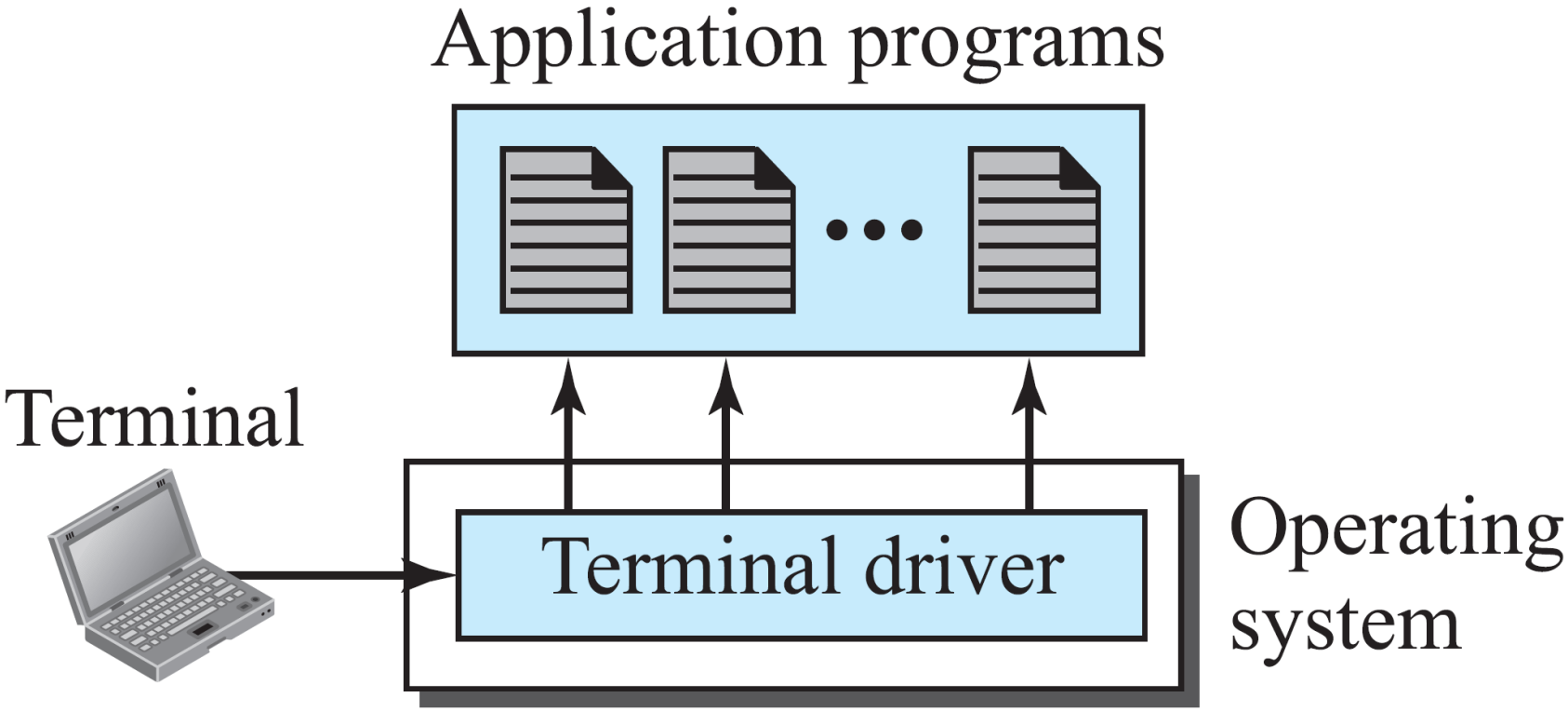
TELNET

- TELNET **requires** a **logging name** and **password**.
- TELNET is **vulnerable** to hacking because it **sends** all data including the **password** in **plaintext** (not encrypted).
- **Because** of this **security issue**, TELNET is almost **replaced** by Secure Shell (SSH).

Why Study TELNET

- The **simple** plaintext architecture of TELNET.
- Network **administrators often** use TELNET for **diagnostic** and **debugging** purposes.

Local Logging

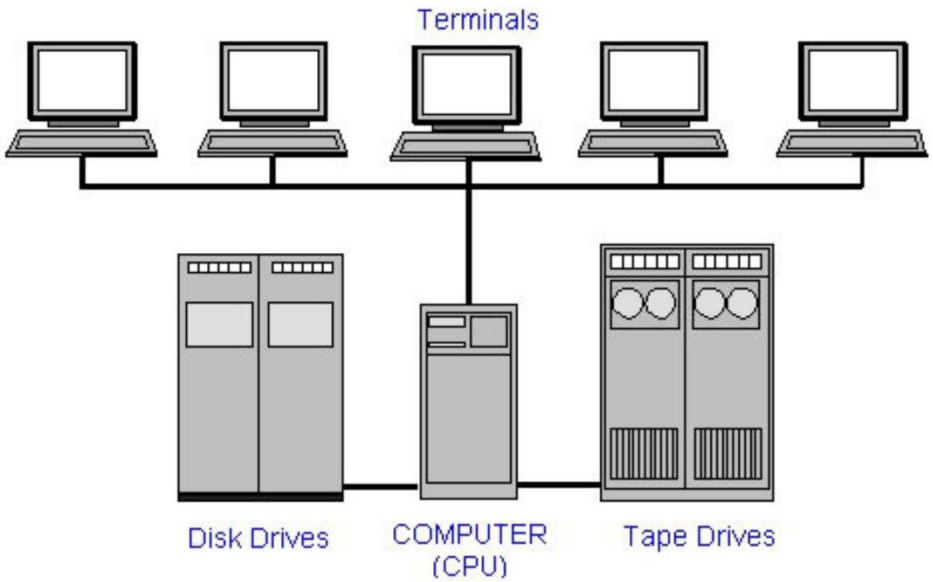


a. Local logging

Local Logging

- As a **user types** at a **terminal** or at a workstation running a **terminal emulator**, the keystrokes are accepted by the terminal driver.
- The terminal driver passes the characters to the operating system.
- The operating system, in turn, interprets the combination of characters and invokes the desired application program or utility.

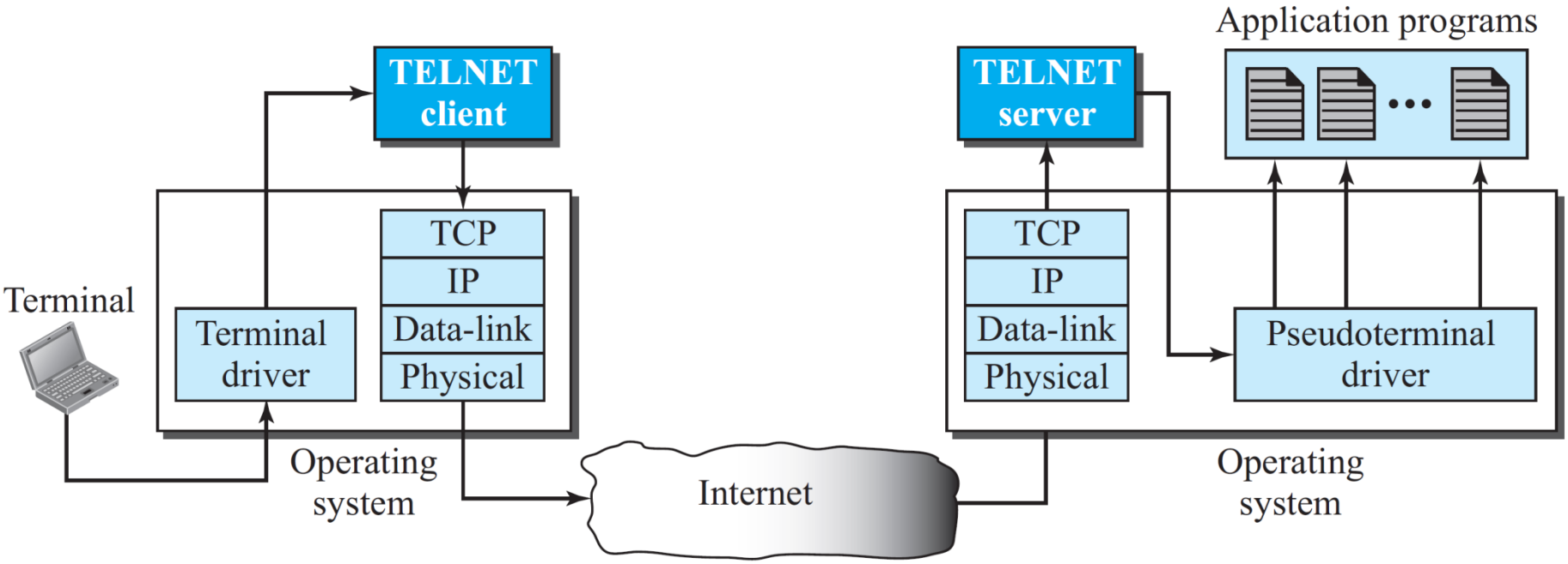
Terminal vs. Workstations



Terminal

A complete Computer Workstation

Remote Logging



b. Remote logging

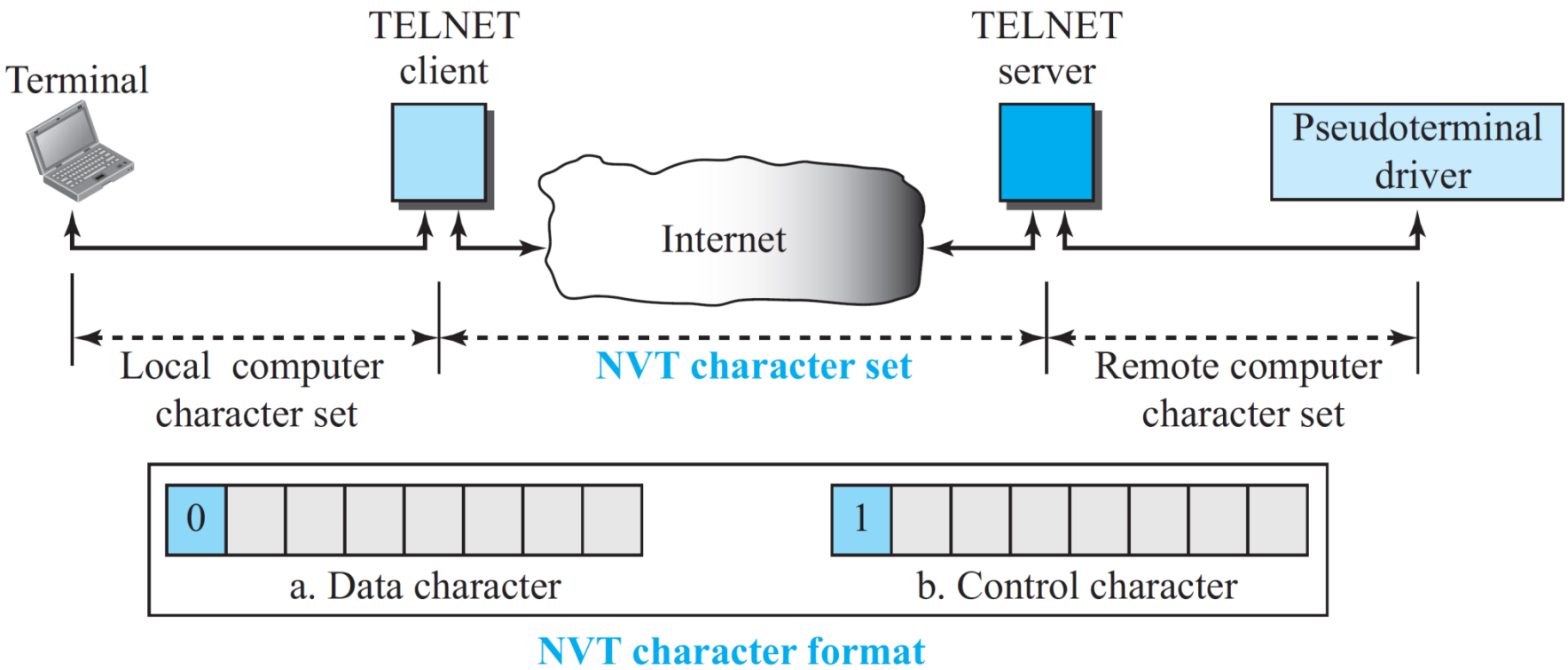
Remote Logging

- **TELNET** user **sends** the **keystrokes** to the **terminal** driver where the **local operating system** accepts the characters but **does not interpret** them.
- **TELNET** client **transforms** the **characters** into a **universal** character set called *Network Virtual Terminal* (**NVT**).
- **TELNET** server **changes** the characters to the **corresponding** characters **understandable** by the **remote** computer.
- ***pseudoterminal*** *driver*, which **pretends** that the characters are **coming** from a **terminal**.

Network Virtual Terminal (NVT)

- Every computer and its operating system **accepts** a special **combination** of characters as **tokens**.
- **For example**,
  - the **end-of-file** token in a computer running the **DOS** operating system is **Ctrl+z**,
  - while the **UNIX** operating system recognizes **Ctrl+d**.

Network Virtual Terminal (NVT)



Examples of TELNET interface commands

Command	*Meaning*	*Command*	*Meaning*
open	Connect to a remote computer	**set**	Set the operating parameters
close	Close the connection	**status**	Display the status information
display	Show the operating parameters	**send**	Send special characters
mode	Change to line or character mode	**quit**	Exit TELNET

Secure Shell (SSH)

- SSH is a **secure** application **program** that can be **used** today for **several purposes** such as **remote logging** and **file transfer**.

- **versions** of SSH: SSH-1 and SSH-2,

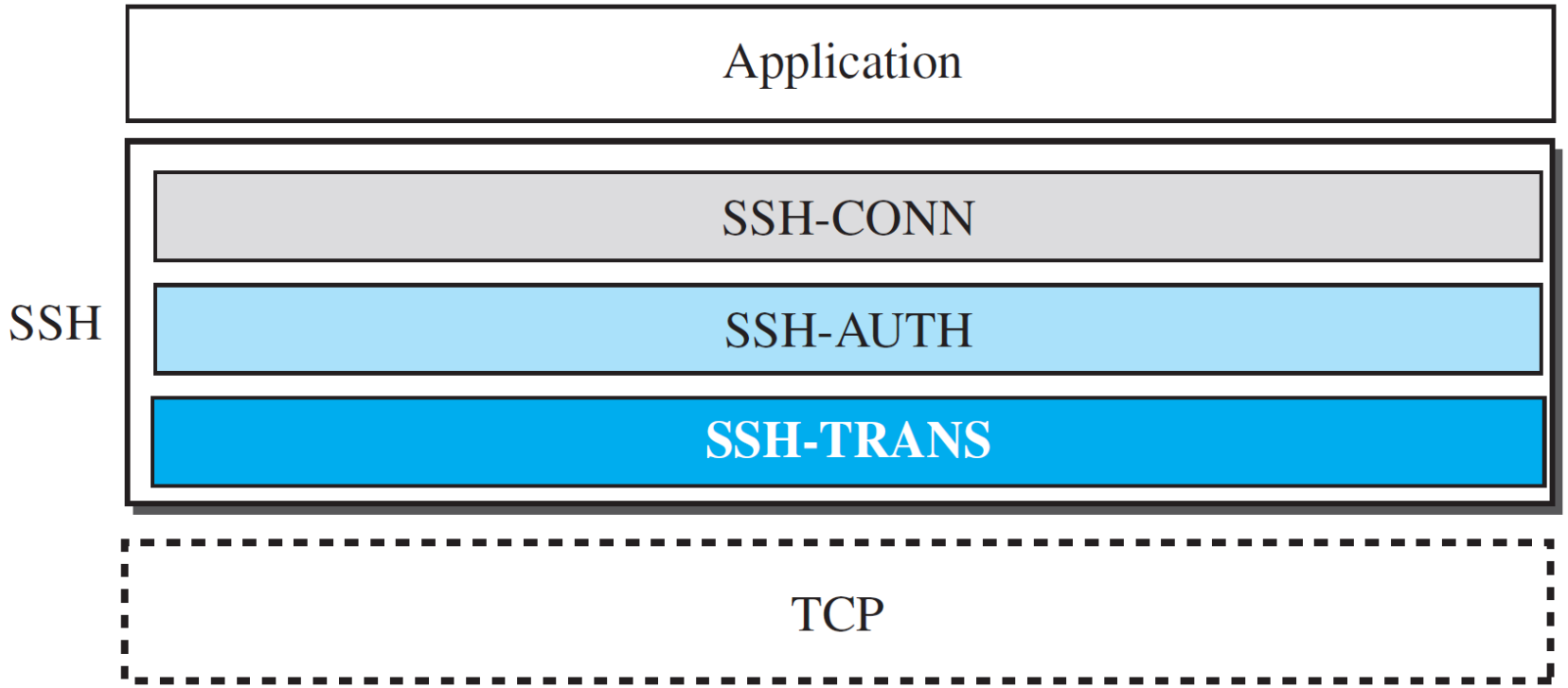
- SSH-1 has **security problems**.

SSH Components

– SSH is an Application-layer protocol with three components:

1. SSH **Transport-Layer** Protocol (SSH-TRANS)
2. SSH **Authentication** Protocol (SSH-AUTH)
3. SSH **Connection** Protocol (SSH-CONN)

SSH Components



SSH-TRANS

- SSH uses **SSL/TLS** protocol that creates a **secured channel** on top of the TCP.
  - Provides **privacy** or **confidentiality** of the message exchanged.
  - Provides data **integrity**, which means that that the messages exchanged are not changed by an intruder.
  - Server **authentication**, which means that the client is now sure that the server is the one that it claims to be.
  - **Compression** of the messages.

SSH-AUTH

- SSH can **call** another procedure (SSL/TLS) that can **authenticate** the client for the server.

- The server **responds** with either:
  - a **success message**, which confirms that the client is authenticated
  - or a **failed message**, which means that the process needs to be repeated with a new request message.

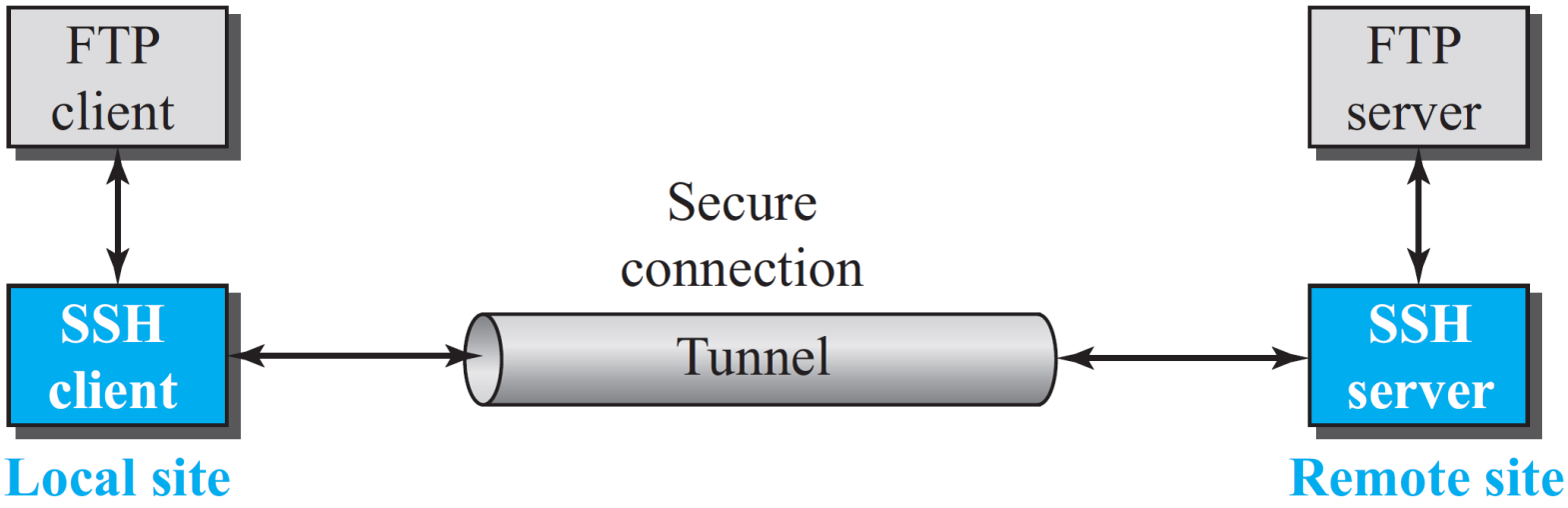
SSH-CONN

- One of the services provided by the SSH-CONN protocol is **multiplexing**.
- SSH-CONN takes the **secure channel** established by the two previous protocols and lets the client create **multiple logical channels** over it.
- **Each channel** can be used for a **different purpose**, such as remote **logging**, **file transfer**, and so on.

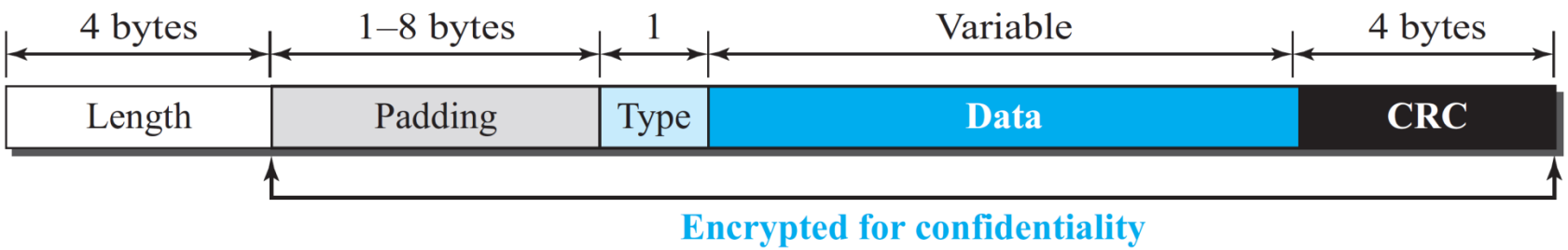
SSH Applications

- ***SSH for Remote Logging***
  - PuTTy, Tectia
- ***SSH for File Transfer***
  - *Secure File Transfer Program* (*sftp*)
  - *Secure Copy* (*scp*)
- ***Port Forwarding (***SSH *tunneling)***
  - uses the **secured channels** available in SSH to access an application program that **does not provide** security services.(such as Telnet and SMTP)

Port Forwarding



Format of the SSH Packets

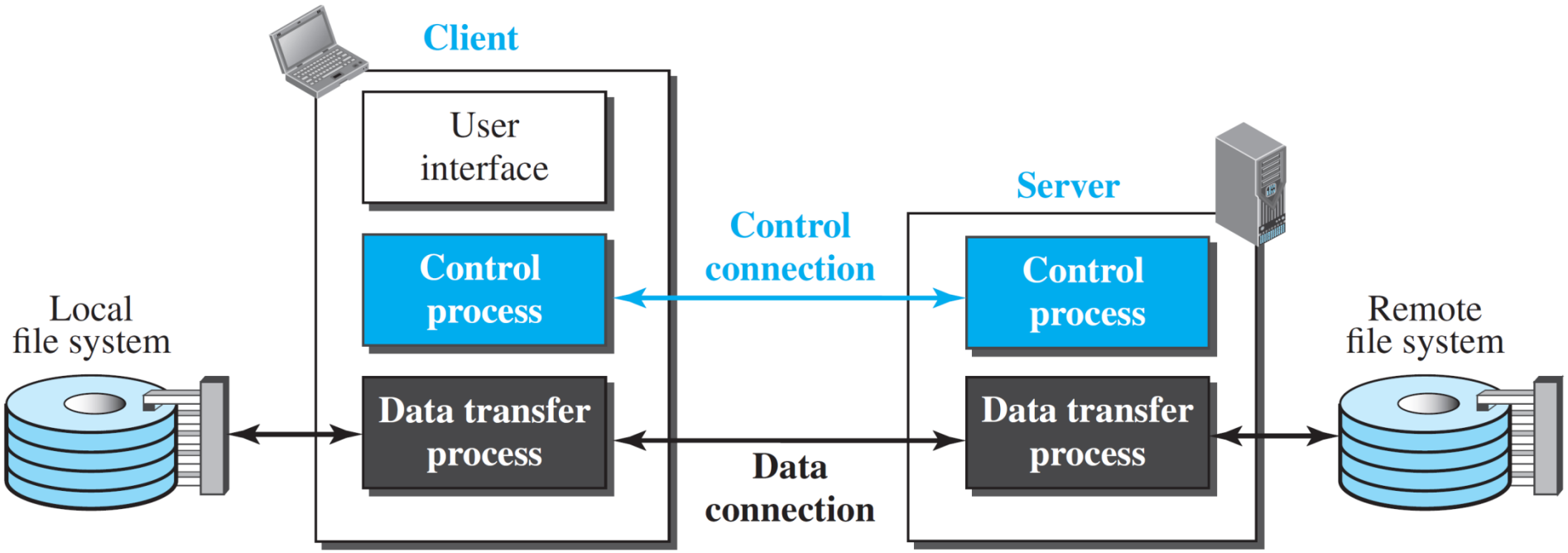


File Transfer Protocol (FTP)

- **FTP** is the **standard** protocol provided by TCP/IP for **copying** a file from **one** host to **another**.

- **Problems** in transferring files:
  - The two systems may use different **file name** conventions.
  - Two systems may have different ways to **represent data**.
  - Two systems may have different **directory structures**.

File Transfer Protocol (FTP)



FTP Two Connections

- Control Connection
  - TCP port 21
- Data Connection
  - TCP port 20

Discussion